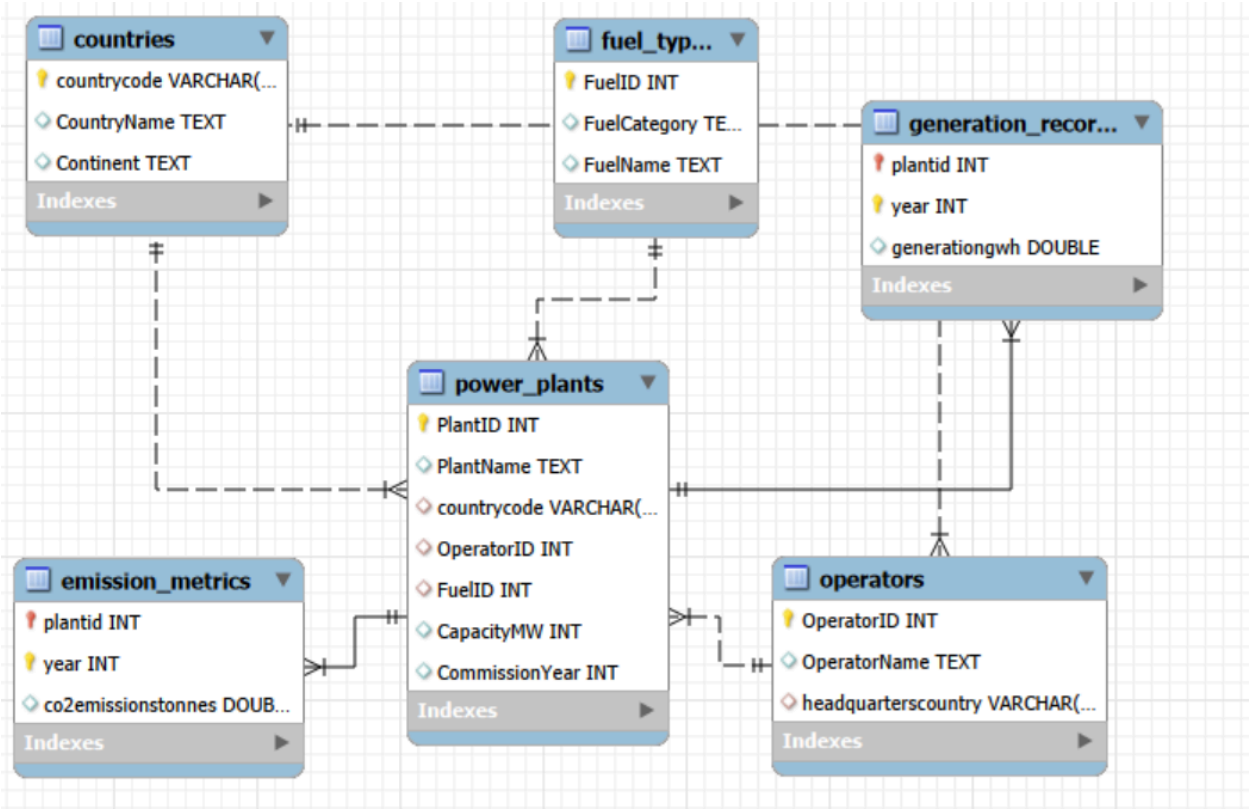
Colin Donahoe
Assignment 3
EER Diagram:



1.

HW1* SQL File 8*

Limit to 1000 rows

```
27  select power_plants.plantname,
28         countries.countryname,
29         operators.operatorname,
30         fuel_types.fuelcategory,
31         fuel_types.fuelname,
32         power_plants.capacitymw,
33         power_plants.commissionyear
34  from power_plants
35  join countries on power_plants.countrycode = countries.countrycode
36  join operators on power_plants.operatorid = operators.operatorid
37  join fuel_types on power_plants.fuelid = fuel_types.fuelid
38  ORDER BY power_plants.capacitymw DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

```
39
40 •  SELECT power_plants.plantname,
41           power_plants.countrycode,
42           generation_records.year,
43           generation_records.generationgwh
44     FROM power_plants
45     JOIN generation_records ON power_plants.plantid = generation_records.plantid
46     WHERE generation_records.year = 2024
47     ORDER BY generation_records.generationgwh DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

plantname	countrycode	year	generationgwh
Three Gorges	CHN	2024	98800
Itaipu	BRA	2024	89000.1
Tuoketuo	CHN	2024	71000.3
Vindhyachal	IND	2024	35600.8
Grand Coulee	USA	2024	21500.5
Janschwalde	DEU	2024	18400.2
Loy Yang	AUS	2024	14200.4
Kashiwazaki-Kariwa	JPN	2024	0

2.

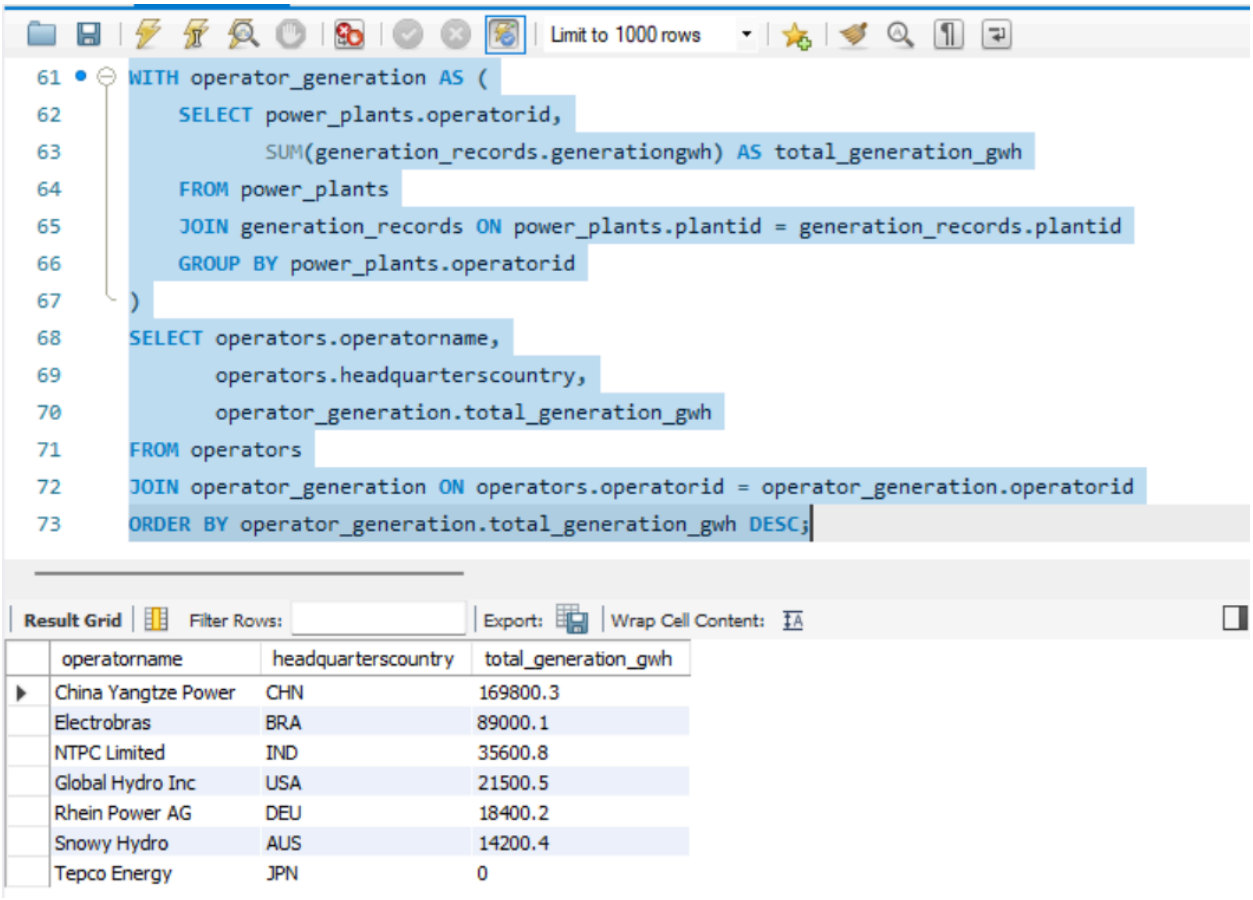
```
48
49 •  SELECT power_plants.plantname,
50           power_plants.countrycode,
51           generation_records.year,
52           generation_records.generationgwh,
53           emission_metrics.co2emissionstonnes
54     FROM power_plants
55     JOIN generation_records ON power_plants.plantid = generation_records.plantid
56     JOIN emission_metrics   ON generation_records.plantid = emission_metrics.plantid
57                            AND generation_records.year    = emission_metrics.year
58     WHERE generation_records.year = 2024
59     ORDER BY emission_metrics.co2emissionstonnes ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

plantname	countrycode	year	generationgwh	co2emissionstonnes
Grand Coulee	USA	2024	21500.5	0
Itaipu	BRA	2024	89000.1	0
Kashiwazaki-Kariwa	JPN	2024	0	0
Three Gorges	CHN	2024	98800	0
Loy Yang	AUS	2024	14200.4	1950000
Janschwalde	DEU	2024	18400.2	2450000
Vindhyachal	IND	2024	35600.8	3100000
Tuoketuo	CHN	2024	71000.3	5200000

3.

4.



```
Limit to 1000 rows
61  WITH operator_generation AS (
62      SELECT power_plants.operatorid,
63             SUM(generation_records.generationgwh) AS total_generation_gwh
64      FROM power_plants
65      JOIN generation_records ON power_plants.plantid = generation_records.plantid
66      GROUP BY power_plants.operatorid
67  )
68  SELECT operators.operatorname,
69         operators.headquarterscountry,
70         operator_generation.total_generation_gwh
71  FROM operators
72  JOIN operator_generation ON operators.operatorid = operator_generation.operatorid
73  ORDER BY operator_generation.total_generation_gwh DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

operatorname	headquarterscountry	total_generation_gwh
China Yangtze Power	CHN	169800.3
Electrobras	BRA	89000.1
NTPC Limited	IND	35600.8
Global Hydro Inc	USA	21500.5
Rhein Power AG	DEU	18400.2
Snowy Hydro	AUS	14200.4
Tepco Energy	JPN	0

5.

```sql
WITH country_generation AS (
    SELECT power_plants.countrycode,
           SUM(generation_records.generationgwh) AS total_generation_gwh
    FROM power_plants
    JOIN generation_records ON power_plants.plantid = generation_records.plantid
    GROUP BY power_plants.countrycode
),
country_emissions AS (
    SELECT power_plants.countrycode,
           SUM(emission_metrics.co2emissionstonnes) AS total_co2_tonnes
    FROM power_plants
    JOIN emission_metrics ON power_plants.plantid = emission_metrics.plantid
    GROUP BY power_plants.countrycode
)
SELECT countries.countryname,
       country_generation.total_generation_gwh,
       country_emissions.total_co2_tonnes
FROM countries
JOIN country_generation ON countries.countrycode = country_generation.countrycode
JOIN country_emissions  ON countries.countrycode = country_emissions.countrycode
ORDER BY country_generation.total_generation_gwh DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

countryname	total_generation_gwh	total_co2_tonnes
China	169800.3	5200000
Brazil	89000.1	0
India	35600.8	3100000
United States	21500.5	0
Germany	18400.2	2450000

Result 6